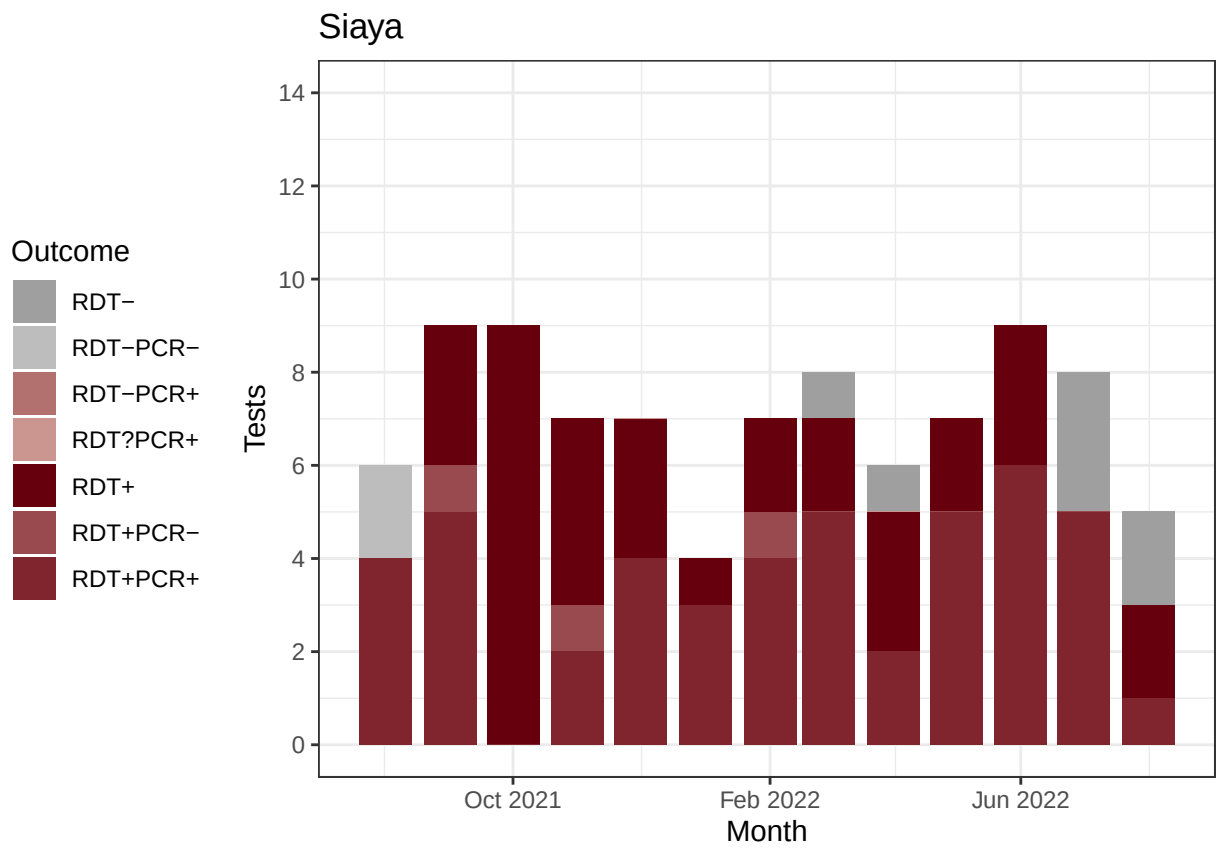
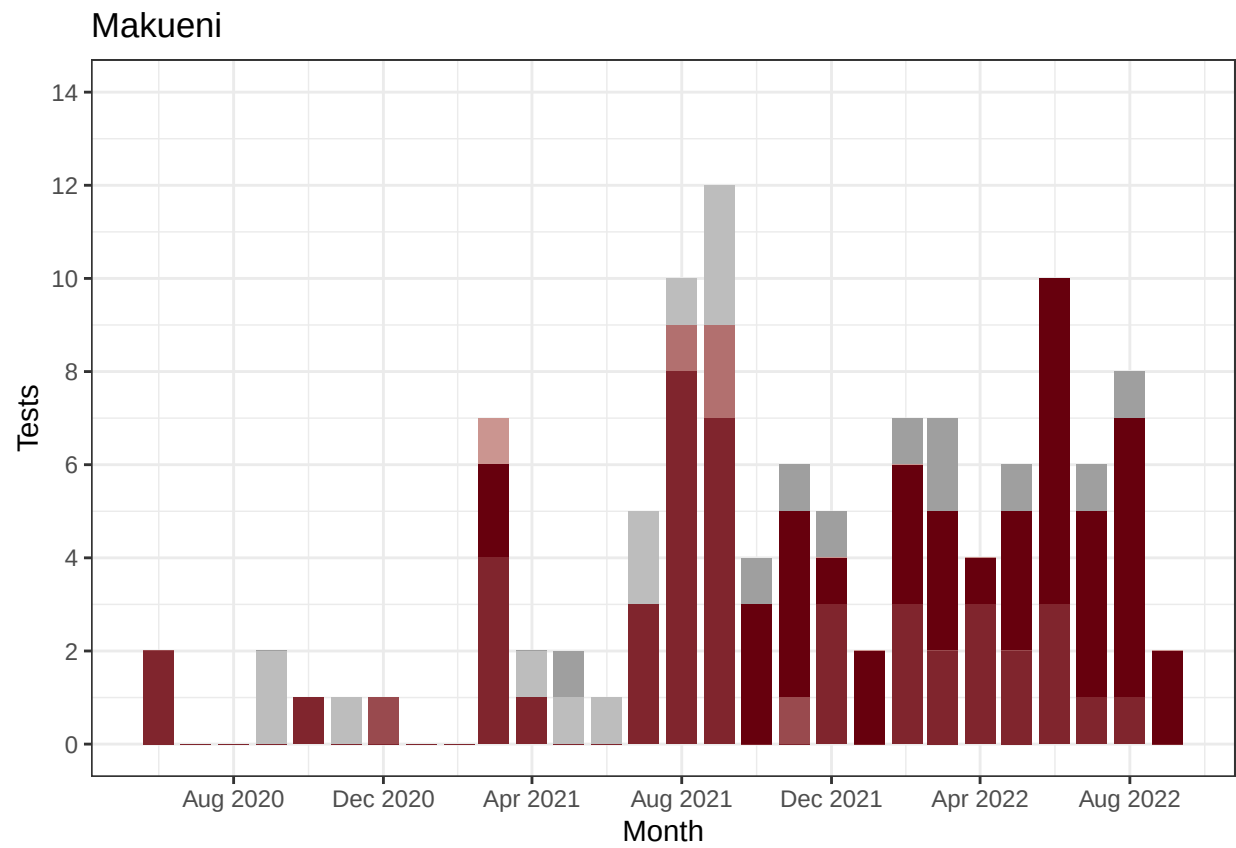


Paper_Se_Sp_analysis

Mumbua Mutunga

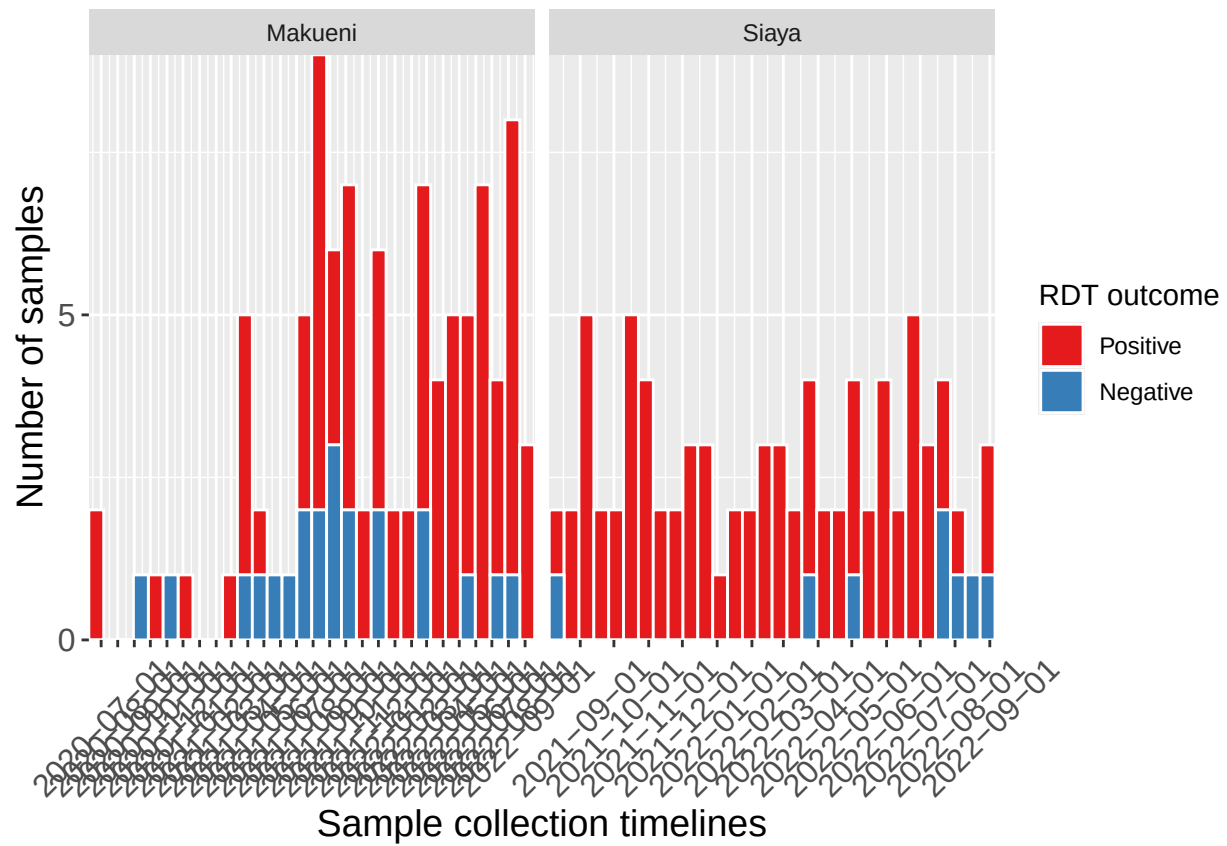
2026-02-25

Time series



```
## PointEst Lower Upper
## 0.9892473 0.806771 1.212946
```

```
## PointEst Lower Upper
## 0.3845494 0.3196306 0.4626473
```



clinical signs

```
# cleaning the contact tracing data
```

```
ct_data <- import(here::here("data", "clean", "ct_data.rds"))
```

```
## Warning: Missing 'trust' will be set to FALSE by default for RDS in 2.0.0.
```

```
ct_data <- import("data/clean/ct_data.rds")
```

```
## Warning: Missing 'trust' will be set to FALSE by default for RDS in 2.0.0.
```

```
# figure using the entire dataset (205 samples)
```

```
Fig <- ct_data |>
```

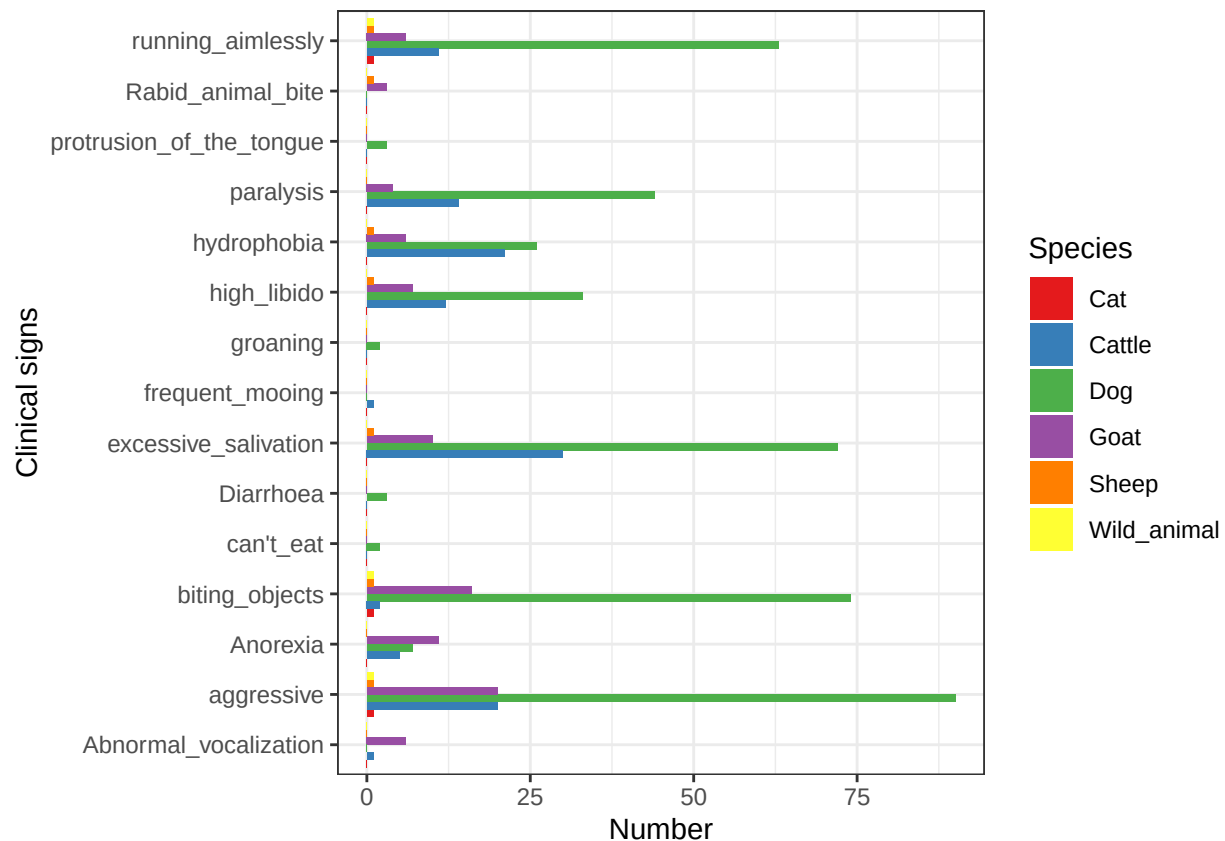
```

select(sample_collection_dt, species, cnty_name, clinical_signs_, lab_diagnostics_pcr, rapid_diagnost.
#filter(!lab_diagnostics_pcr %in% "") |>
group_by(species, clinical_signs_) |>
dplyr::count()

# Revise so as to capture - bc this should be per dog ID (not repeating)
Fig_new = Fig
for(i in unique(Fig$clinical_signs_)){
  for(j in unique(Fig$species)){
    if(nrow(Fig[Fig$clinical_signs_ == i &
               Fig$species == j,]) == 0){
      new_row = data.frame(
        clinical_signs_ = i,
        species = j,
        n = 0)
      Fig_new = rbind(Fig_new, new_row)
    }
  }
}

ggplot(data = Fig_new)+
  geom_col(aes(y=n, x = clinical_signs_, fill = species), position = "dodge") +
  labs(x = "Clinical signs", y = "Number", fill="Species")+
  scale_fill_brewer(palette = "Set1", direction = 1)+
  coord_flip()+
  theme_bw()

```



```
ggsave("Figs/Figure_4_supplementary.png",
#ggsave("/Users/mumbuamutunga/Library/CloudStorage/Dropbox/Desktop/Sp:Se_Analysis/Sp_Se-analysis/Figs/F
width = 18, height = 12)

# figure using the sequenced samples

Fig_seq <- ct_data |>
  select(sample_collection_dt, species, cnty_name, clinical_signs_, lab_diagnostics_pcr, rapid_diagnost
  filter(!lab_diagnostics_pcr %in% "") |>
  group_by(species, clinical_signs_) |>
  dplyr::count()

Fig_new_seq <- Fig_seq

for(i in unique(Fig_seq$clinical_signs_)){
  for(j in unique(Fig_seq$species)){
    if(nrow(Fig_seq[Fig_seq$clinical_signs_ == i &
      Fig_seq$species == j,]) == 0){
      new_row = data.frame(
        clinical_signs_ = i,
        species = j,
        n = 0)
      Fig_new_seq = rbind(Fig_new_seq, new_row)
    }
  }
}
```

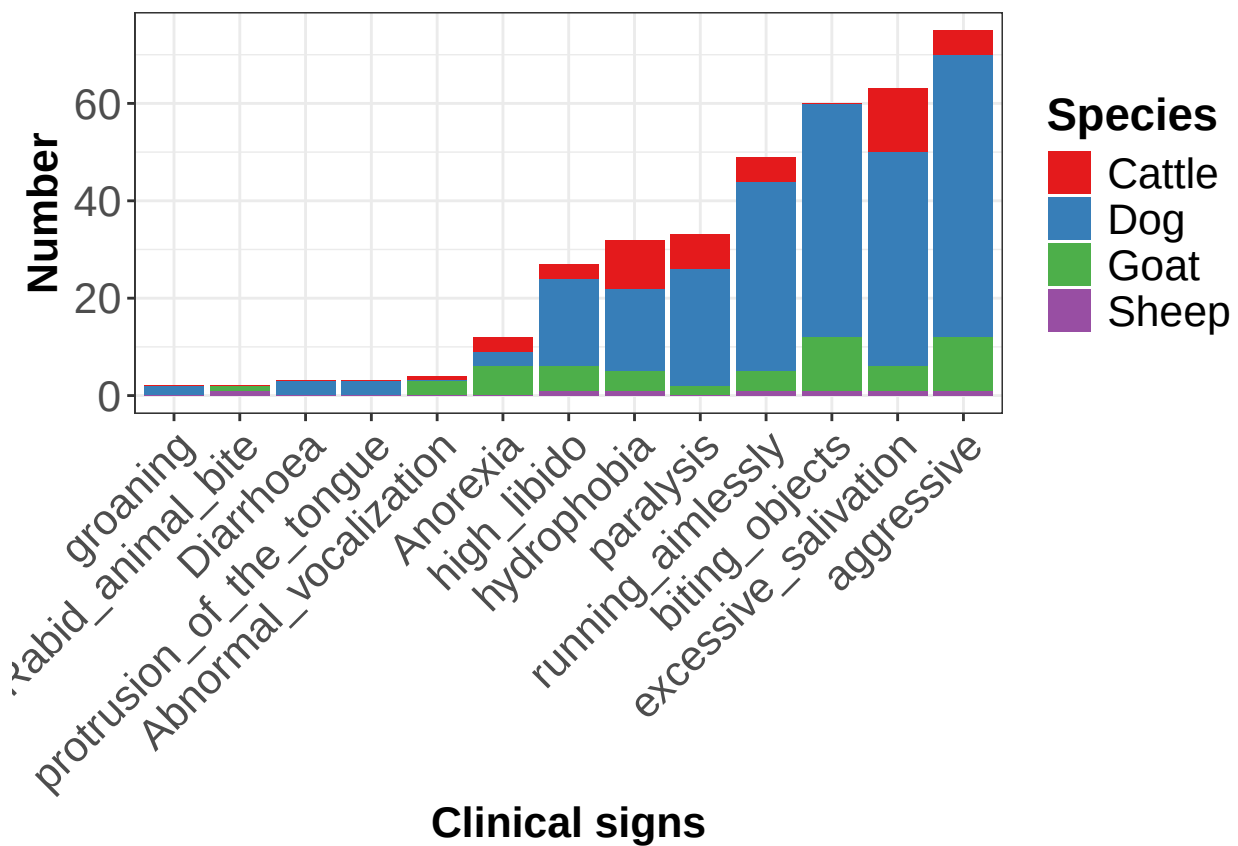
```

}

# Reorder clinical_signs_ by total frequency (n) in ascending order
Fig_new_seq <- Fig_new_seq %>%
  group_by(clinical_signs_) %>%
  mutate(total_n = sum(n)) %>%
  ungroup() %>%
  mutate(clinical_signs_ = reorder(clinical_signs_, total_n))

ggplot(data = Fig_new_seq) +
  geom_col(aes(x = clinical_signs_, y = n, fill = species), position = "stack") +
  labs(x = "Clinical signs", y = "Number", fill = "Species") +
  scale_fill_brewer(palette = "Set1", direction = 1) +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 16),
    axis.text.y = element_text(size = 16),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    legend.text = element_text(size = 16),
    legend.title = element_text(size = 17, face = "bold")
  )

```



```

ggsave("Figs/Figure-4.png",
# ggsave("/Users/mumbuamutunga/Library/CloudStorage/Dropbox/Desktop/Sp:Se_Analysis/Sp_Se-analysis/Figs/

```

```

width = 15, height = 10)

#incidence of positive cases per year

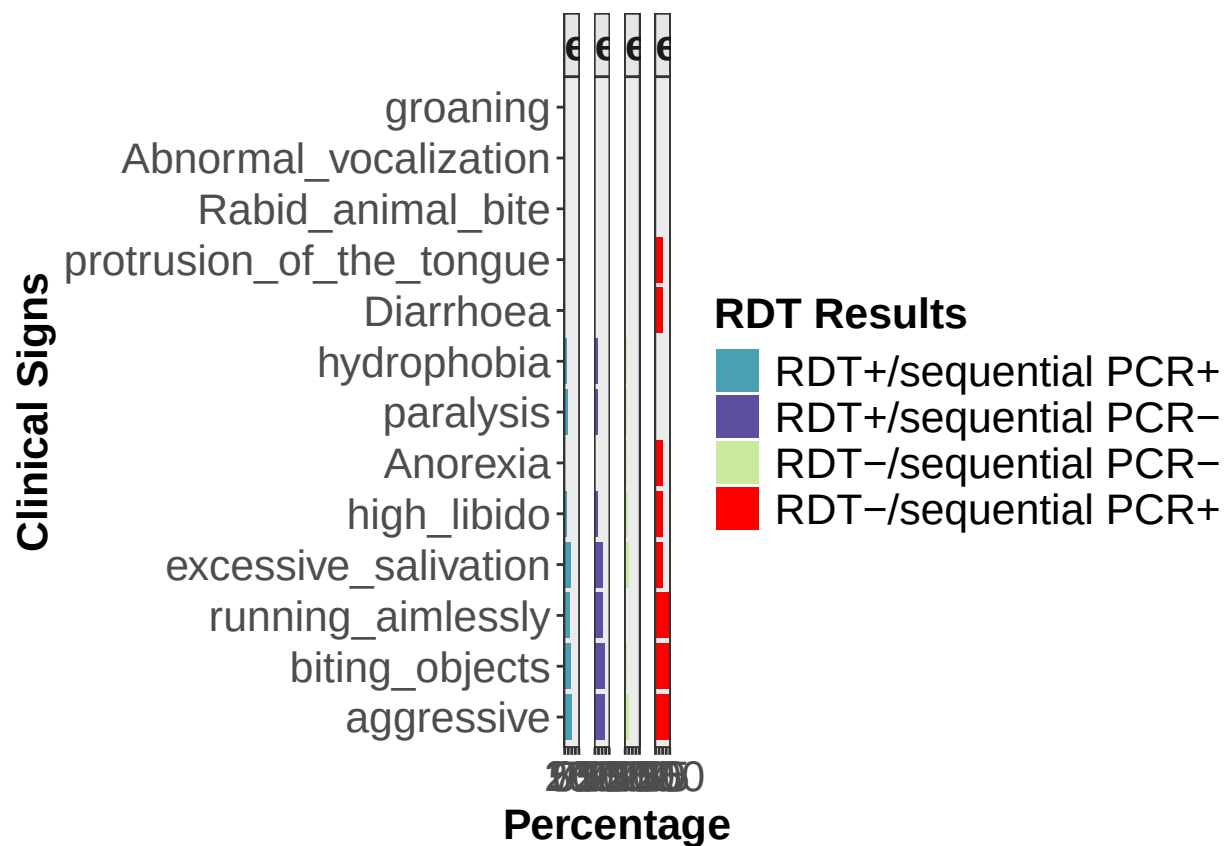
inc_mak <- combined |>
  select(date_collected, county, rapid_diagnostic_test) |>
  filter(rapid_diagnostic_test %in% "Positive") |>
  filter(county %in% "Makueni") |>
  mutate(year = format(date_collected, "%Y")) |> # Extract the year
  group_by(year) |> # Group by year
  count()

inc_siaya <- combined |>
  filter(rapid_diagnostic_test %in% "Positive") |>
  filter(county %in% "Siaya") |>
  mutate(year = format(date_collected, "%Y")) |> # Extract the year
  group_by(year) |> # Group by year
  count()

# if in two counties we have 94 samples that are positive if we could escalate this to the remaining 45

```

data with RDT & clinical signs

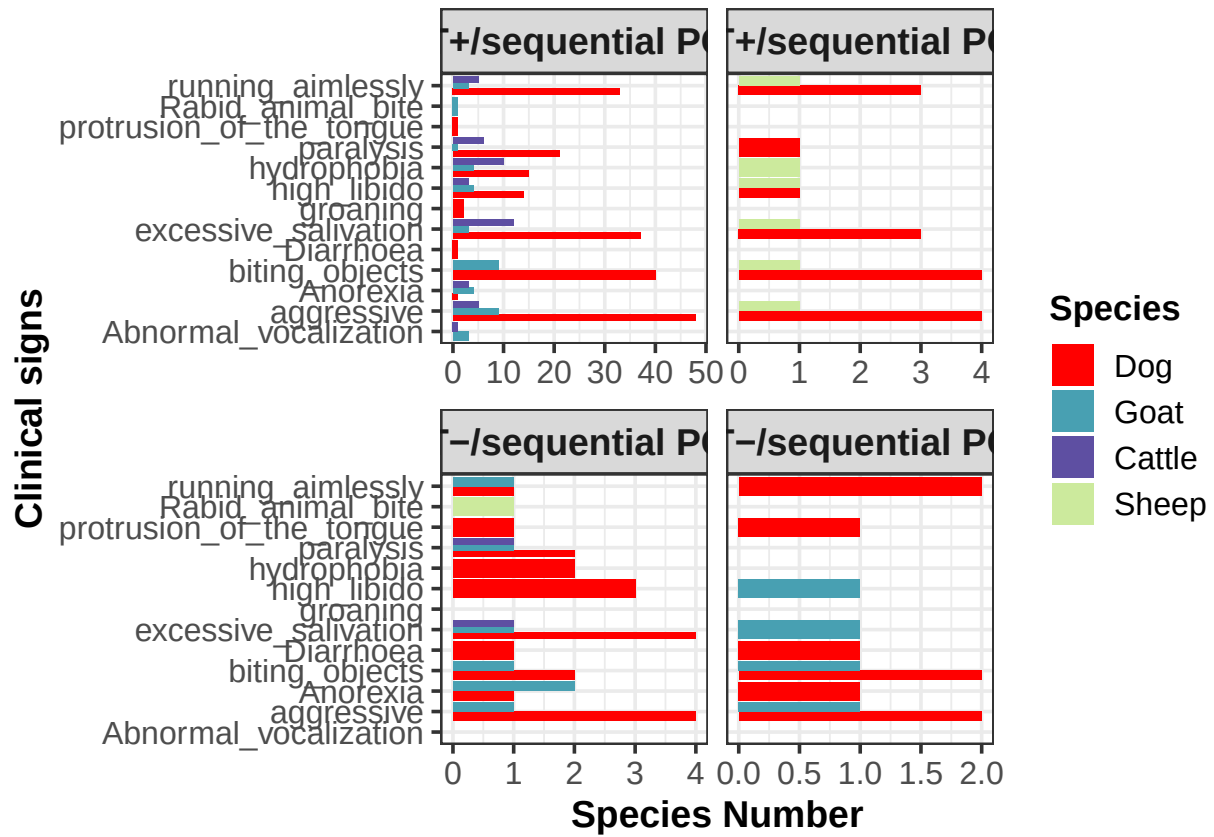


#Combining figure 6 and 8 into one

filter out zeros from the dataset

```
new_plan_zeros <- new_plan_zeros |>
  filter(n != 0)
```

```
ggplot(new_plan_zeros, aes(x = n,
                           y = clinical_signs_,
                           fill = species)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("red", "#48A0B2", "#5E4FA2", "#CCEA9D")) +
  facet_wrap(~ test_result, scales = "free_x") +
  labs(x = "Species Number",
       y = "Clinical signs",
       fill = "Species") +
  theme_bw(base_size = 14) +
  theme(
    strip.text = element_text(size = 14, face = "bold"),
    axis.text.x = element_text(size = 12),
    axis.text.y = element_text(size = 12),
    axis.title = element_text(size = 14, face = "bold"),
    legend.text = element_text(size = 12),
    legend.title = element_text(size = 13, face = "bold")
  )
```

```
ggsave("Figs/Figure_5.png", width = 18, height = 12)
```

```
#ggsave("/Users/mumbumutunga/Library/CloudStorage/Dropbox/Desktop/Sp:Se_Analysis/Sp_Se-analysis/Figs/F
```